



Alberta Public Health Disease Management Guidelines

Cytomegalovirus, Congenital (CMV)



This publication is issued under the Open Government Licence – Alberta (<http://open.alberta.ca/licence>). Please note that the terms of this licence do not apply to any third-party materials included in this publication.

This publication is available online at <https://open.alberta.ca/publications/congenital-cytomegalovirus-cmv>

For further information on the use of this guideline contact:

Health.CD@gov.ab.ca

Communicable Disease Control Branch

Public Health Division

Alberta Health

Cytomegalovirus, Congenital (CMV) | Alberta Health, Government of Alberta

© 2024 Government of Alberta | November 2024



Table of Contents

Revisions	4
Case Definition	5
Confirmed Case	5
Probable Case	5
Reporting Requirements	6
Physicians, Health Practitioners and Others	6
Laboratories	6
Alberta Health Services and First Nations Inuit Health Branch	6
Clinical Assessment and Epidemiology	7
Etiology	7
Clinical Presentation.....	7
Diagnosis	7
Treatment.....	7
Reservoir.....	7
Transmission.....	7
Incubation Period	7
Period of Communicability.....	7
Host Susceptibility	8
Incidence.....	8
Public Health Management.....	9
Key Investigation.....	9
Management of a Case.....	9
Management of Contacts	9
Preventive Measures.....	9
References	10

Revisions

Revision Date	Document Section	Description of Revision
November 2024	General	Entire Guideline reviewed and updated to align with current information
	Case Definition	Changed from two to three week window post birth for case definition
	Reporting Requirements	Added link to “Perinatally Acquired Notifiable Disease Enhanced Report Form”
	Clinical Presentation	Updated risk estimates based upon current literature
	Incidence	Added link to Interactive Health Data Application (IHDA) for access to current data

Case Definition

Confirmed Case

Laboratory confirmation of cytomegalovirus (CMV) infection in a clinical specimen collected within the first three weeks of life with or without clinical illness^(A):

- Detection of CMV virus from an appropriate clinical specimen^(B) by molecular diagnostic techniques (i.e. quantitative PCR),

OR

- Histopathological evidence of CMV inclusion disease from an appropriate clinical specimen^(B).

Probable Case

Clinical illness ^(A) in a child/infant of any age, born to a CMV seropositive childbearing parent.

^(A) See [Clinical presentation](#)

^(B) Refer to the [Public Health Laboratory \(ProvLab\) Guide to Services](#) for current specimen collection and submission information.

Reporting Requirements

Physicians, Health Practitioners and Others

Physicians, health practitioners and others shall notify the Zone Medical Officer of Health (MOH) (or designate), of all confirmed and probable cases in the prescribed form by mail, fax or electronic submission within 48 hours (two business days).

Laboratories

All laboratories shall report all positive laboratory results by mail, fax or electronic submission within 48 hours (two business days) to the:

- Chief Medical Officer of Health (CMOH) (or designate), and
- Zone MOH (or designate).

Alberta Health Services and First Nations Inuit Health Branch

- The Zone MOH (or designate) where the case currently resides shall forward the [Perinately Acquired Notifiable Disease Enhanced Report form](#) for all confirmed and probable congenital cases to the CMOH (or designate) within four weeks of notification.
- For out-of-province and out-of-country cases, the Zone MOH should forward the following information to the CMOH (or designate) by phone, fax or electronic submission within 48 hours (two business days):
 - name,
 - date of birth,
 - out-of-province health care number,
 - out-of-province address and phone number,
 - positive laboratory report, and
 - other relevant clinical/epidemiological information.

Clinical Assessment and Epidemiology

Etiology

Cytomegalovirus (CMV) is a slowly replicating DNA virus. It is a member of the *Herpesvirus* group (herpesvirus 5).^(1–3)

Clinical Presentation

Approximately 1% of all newborns are perinatally infected with CMV, making it the most common congenital infection.^(3–5) For those born with CMV, approximately 10–15% of newborns demonstrate symptoms of infection including, intrauterine growth restriction, neonatal jaundice, purpura, hepatosplenomegaly, microcephaly, brain damage, intracerebral calcifications, and retinitis.^(3,6) However, as many as 6–25% of those born infected but asymptomatic can develop complications including permanent sensorineural hearing loss (SNHL) and psychomotor mental delay.^(7,8) Death may occur in utero.^(1,3) Post-natal morbidity occurs in 30–40% of symptomatic cases.⁽⁶⁾

Diagnosis

The diagnosis of confirmed cases is by detection of the virus from [an appropriate clinical specimen](#).^{(B)(8,9)}

Treatment

- Supportive treatment.
- There is no cure for congenital CMV although treatment with antivirals could offer some hearing improvement and reduce developmental delays if used during the first year of life for symptomatic cases.^(1–3)

Reservoir

Humans.^(3,10)

Transmission

The fetus may be infected in utero from either a primary or reactivated parental infection with the former having 10-15 times higher transmission rates.^(1,8) Congenital infections occurs by vertical transmission of CMV to fetus in utero by transplacental passage of parental virus.^(6,10) Children born with CMV may shed virus for five to six years and have potential to infect others.⁽³⁾

Incubation Period

Incubation period varies with the route of infection (i.e., congenital vs perinatal) and the size of inoculum.⁽⁹⁾

Period of Communicability

The virus is excreted in the urine and saliva of newborns for many months. The excretion may persist or be episodic for several years following a primary infection, possibly five or six years following a neonatal infection. Those infected with CMV generally begin shedding and can transmit virus approximately one month after becoming infected.⁽³⁾

Host Susceptibility

The virus is ubiquitous, thus infection is present in all populations, with sero-prevalence increasing with age and decreasing in developed countries.^(3,9,11) Fetuses; individuals with debilitating diseases, immunodeficiency or immunosuppression; organ recipients and people living with untreated HIV are more susceptible to disease.⁽³⁾ Antibodies develop with infection, but are insufficient to confer immunity, and reactivation can occur after primary infection.^(6,11)

Incidence

Congenital CMV is not notifiable nationally. However, Canadian incidence has been estimated to affect 0.5% to 1% of live births.^(1,6,12) Congenital CMV became notifiable in Alberta in 1985. Refer to the [Interactive Health Data Application \(IHDA\)](#) for more information on disease incidence in Alberta.

Public Health Management

Key Investigation

- Assess whether infant meets case definition for confirmed or probable congenital infection.
- Determine current status of the infant, including presence/absence of symptoms.
- Identify if disease or antibodies are present in birth parent.

Management of a Case

- Supportive.
- For hospitalized cases, follow recommended hospital infection prevention and control precautions.
- Recommend infected infants receive close medical follow up by a pediatrician to monitor for hearing developmental delays which would benefit from intervention.^(1,2,4)
- Additional investigations (bloodwork and imaging) may be warranted to identify evidence of other end organ disease.⁽⁵⁾

Management of Contacts

- Due to the high prevalence of asymptomatic shedders in the population, there is no follow up of contacts.

Preventive Measures

- Pregnant persons should avoid sharing food and utensils/cups with children.^(4,10,13)
- Persons of childbearing potential that work in the following occupational settings should follow routine infection prevention and control practices.
 - hospitals, particularly labour and delivery and pediatric units.
 - daycare centres, with preschoolers or with persons unable to maintain their own personal hygiene.
 - Strict hand hygiene after contact is vital (notably after changing diapers or assisting with toileting)
 - Risk seems to be greatest when caring for children less than two years of age.^(4,10,14)

References

1. Barton M, Forrester AM, McDonald J. Update on congenital cytomegalovirus infection: Prenatal prevention, newborn diagnosis, and management. *Paediatr Child Heal* [Internet]. 2020 [cited 2024 Oct 15];25(6):395. Available from: <https://cps.ca/documents/position/update-on-congenital-cytomegalovirus-infection-prenatal-prevention-newborn-diagnosis-and-management>
2. Gantt S, Brophy J, Dunn J, Vaudry W, Bitnun A, Renaud C, et al. AMMI Canada: Response to FAQs about the management of children with congenital cytomegalovirus infection in Canada. *J Assoc Med Microbiol Infect Dis Canada* [Internet]. 2019;4(4):208–14. Available from: <https://pmc.ncbi.nlm.nih.gov/articles/PMC9612812/>
3. Lanzieri, T., Bialek S. Cytomegalovirus Disease. In: Heymann DL (ed), editor. *Control of Communicable Diseases*. 21st Edict. Washington D.C.: American Public Health Association; 2022. p. 143–6.
4. Pesch MH, Kuboushek K, Mckee MM, Thorne MC, Weinberg JB. Congenital cytomegalovirus infection. *BMJ* [Internet]. 2021; Available from: <http://dx.doi.org/10.1136/bmj.n1212>
5. Lim Y, Lyall H. Congenital cytomegalovirus – who, when, what-with and why to treat? *J Infect* [Internet]. Elsevier Ltd; 2017;74:S89–94. Available from: [http://dx.doi.org/10.1016/S0163-4453\(17\)30197-4](http://dx.doi.org/10.1016/S0163-4453(17)30197-4)
6. Tesini BL. Congenital and Perinatal Cytomegalovirus infection (CMV) [Internet]. *Merck Manual (Professional Version)*. 2022 [cited 2024 Oct 15]. Available from: <https://www.merckmanuals.com/en-ca/professional/pediatrics/infections-in-neonates/congenital-and-perinatal-cytomegalovirus-infection-cmv>
7. Government of Canada. Pathogen Safety Data Sheets: Infectious Substances - Cytomegalovirus [Internet]. 2011 [cited 2023 Feb 15]. Available from: <https://www.canada.ca/en/public-health/services/laboratory-biosafety-biosecurity/pathogen-safety-data-sheets-risk-assessment/cytomegalovirus.html>
8. Dietrich, M. & Schieffelin J. Congenital Cytomegalovirus Infection. *Ochsner J* [Internet]. 2019;12(2):123–30. Available from: <https://pmc.ncbi.nlm.nih.gov/articles/PMC6584201/>
9. Committee on Infectious Diseases AA of P. Cytomegalovirus Infection. In: D Kimberlin et al, editor. *Red Book: 2021–2024 Report of the Committee on Infectious Diseases*. 32nd ed. Itasca, IL: America Academy of Pediatrics; 2021. p. 294–300.
10. Akpan US, Pillarisetty LS. Congenital Cytomegalovirus Infection. *StatPearls* [Internet]. StatPearls Publishing; 2021 Aug 11; Available from: www.ncbi.nlm.nih.gov/books/NBK541003/
11. Fowler K, Mucha J, Neumann M, Lewandowski W, Kaczanowska M, Gryś M, et al. A systematic literature review of the global seroprevalence of cytomegalovirus: possible implications for treatment, screening, and vaccine development. *BMC Public Health* [Internet]. BioMed Central; 2022;22(1):1–15. Available from: <https://doi.org/10.1186/s12889-022-13971-7>
12. Vaudry W, Lee BE, Rosychuk RJ. Congenital cytomegalovirus infection in Canada: Active surveillance for cases diagnosed by paediatricians. *Paediatr Child Heal*. 2014;19(1):1–5.
13. Centers for Disease Control and Prevention. Clinical Overview of CMV and Congenital CMV [Internet]. [cited 2024 Oct 15]. Available from: <https://www.cdc.gov/cytomegalovirus/hcp/clinical-overview/index.html>
14. Joseph A, Mahida N, Clark G, Irving W, Soo S. Congenital cytomegalovirus infection. *Paediatr Child Heal (United Kingdom)* [Internet]. Elsevier Ltd; 2018;28(6):277–81. Available from: <https://doi.org/10.1016/j.paed.2018.04.005>